1. Final Proposed Method

M3 Research Project · Detection-Constrained Power Allocation for Cognitive ISAC Relays under Imperfect CSI (NCC 2027 submission)

Source: submitted NCC 2027 paper, Sections II-IV and Algorithm 1. Fig. 1 of the paper is in this folder (`Fig1_system_model.png`).

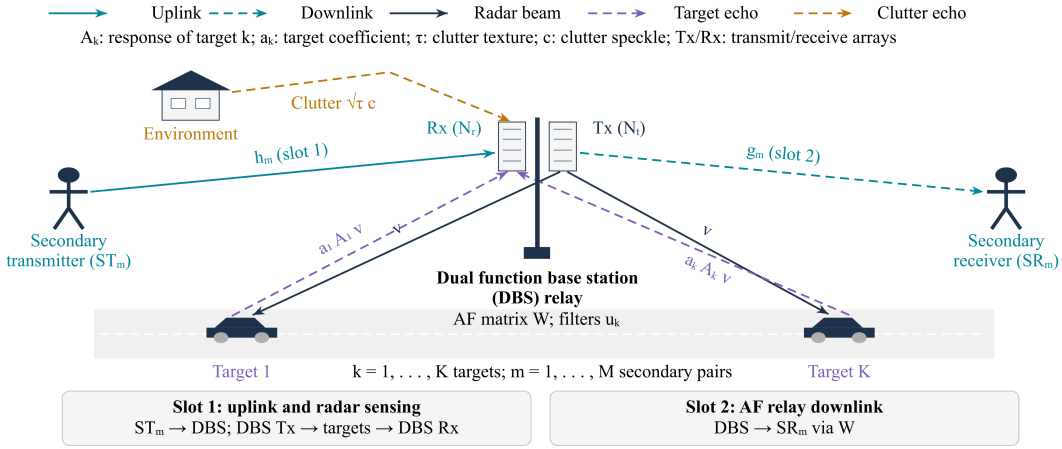


Fig. 1. Cognitive ISAC relay network. Slot 1 combines secondary uplink reception and radar sensing; slot 2 forwards the received signals to the secondary receivers.

System model (Sec. II)

- **Network.** A two-slot amplify-and-forward cognitive ISAC relay network.
- **Base station.** A dual-function base station (DBS) with N_t transmit and N_r receive antennas senses K point targets. It also serves M single-antenna secondary transmitter-receiver (ST-SR) pairs.
- **Slot 1.** The STs transmit while the DBS radiates the radar beam. Radar processing uses this slot.
- **Slot 2.** The DBS relays the received signals to the SRs through the relay matrix W.
- **Disturbance.** Compound-Gaussian clutter, with texture shape ν.
- **CSI.** The uplink and downlink estimates have Gaussian errors. The CSI-error ratio is ε.

Detection law and secondary rate (Sec. III)

- Each target k uses a unit-norm radar filter u_k and a threshold test.
- Leakage from the other targets is modelled as Gaussian interference. This is a surrogate approximation.
- The false-alarm probability P_FA,k (eq. 5) and the detection probability P_D,k (eq. 6) are averaged over the clutter texture and the channel errors.
- The threshold t_k(p) is calibrated so that P_FA,k = α.
- With Gaussian clutter and perfect CSI, P_D >= β reduces to an SINR floor (eq. 7). For $\alpha = 10^{-3}$ and β = 0.9, this floor is 9.80 dB.
- The secondary sum achievable rate (SAR) is R(p, W) (eq. 9). The factor 1/2 accounts for the two slots.

Optimisation problem (eq. 10)

Maximise R(p, W) over the ST powers p, the relay matrix W and the radar filters {u_k}, subject to:

- $0 <= p_m <= P_c$ for every ST,
- relay power $P_t(p, W) <= P_{max}$,
- $||u_k|| = 1$,
- **$P_{D,k}(t_k(p), p) >= \beta$ for every target k**: the detection constraint that replaces the sensing-SINR floor of [3].

Algorithm 1: detection-constrained cognitive ISAC relay design (Sec. IV)

1. Set p to the largest scaling of $P_c \cdot 1$ that meets the detection constraints.
2. Initialise W as a maximal-ratio relay, rescaled by eq. (14), followed by four WMMSE steps.
3. Repeat steps 4-8:
    4. Update each normalised MVDR radar filter (eq. 13). The CSI-error variances provide diagonal loading, so the filter accounts for CSI errors.
    5. Set d = p. Find each directional detection boundary λ_k by bisection (eq. 11, Proposition 1).
    6. Form the linear power surrogates (eq. 12).
    7. Update p by sequential quadratic programming (SQP), then rescale W by (14).
    8. Perform one WMMSE update of W, then rescale by (14).
9. Stop when the relative rate change is below 10^{-4}, or after 40 iterations.
10. Recompute the filters and evaluate eqs. (5)-(6).
11. If a detection constraint fails, shrink p by bisection until every target meets the requirement.
12. Return p, W and the filters.

Numerical details.

- Gauss-Laguerre quadrature over the clutter texture.
- Sobol samples for the uplink errors.
- A design margin of 0.003 on β.

Stated limits.

- The linear surrogates are not global feasibility certificates.
- No global optimality or full-iteration monotonicity is claimed.
- The feasibility check is numerical and applies to the adopted detection surrogate.